	ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)	**Assignment**

Branch	**All**	Programme	**B.Tech**
Course Name	**Mathematics in India**	Semester	**VIII**
Course Code	**IKS 4001**	Academic Year	**2025-2026**
Assignment- 3	**Topic: Ganitapada, Mathematics moves to South, Nila Mathematics**		

Learning Level (LL)	**L1**: Remembering	**L3:** Applying	**L5**: Evaluating
	L2: Understanding	**L4**: Analysing	**L6**: Creating

Q's	Questions	Cos	LL
1	Using Kuttaka method (Euclidean Algorithm) find out a solution to the Diophantine equation: $-13x + 60y = 1$	**CO5**	**L4**
2	What do you mean by the Kerala School of Mathematics? Discuss why this has a significant role in shaping the foundation of modern mathematics in India.	**CO6**	**L2**
3	Briefly describe the geographical setting of the Kerala School of Mathematics in the southern part of India.	**CO6**	**L2**
4	Discuss the key contributions of Madhavadeva of Sangamagrama in the area of algebra and calculus during the 14th century.	**CO6**	**L2**
5	Who was Nilakantha Somayaji? How did he contribute in mathematics and astronomy through his writings?	**CO6**	**L2**
6	Discuss the importance of 'Yuktibhasha' for the study of mathematics and astronomy in the Nila School of Mathematics.	**CO6**	**L2**
7	Briefly elaborate the structural composition of the book 'Yuktibhasha'.	**CO6**	**L2**
8	Give an example of a 4 × 4 magic square, whose magic sum is 40.	**CO6**	**L3**
9	Find the value of x, for which the following 3 × 3 matrix becomes a magic square: $$\begin{bmatrix} x-1 & & x-3 \\ & 6 & \\ x+3 & & x+1 \end{bmatrix}$$ Hence, complete the magic square. You may use the fact that in any 3×3 magic square, the magic sum is always exactly three times the middle element.	**CO6**	**L4**
10	Start from the derivation and then compute the first order interpolating approximations of $\sin(60^{\circ} + \epsilon)$ and $\cos(90^{\circ} + \epsilon)$ for $\epsilon = 10^{-2}$ up to 4 digits after decimal.	**CO6**	**L4**
11	Start from the derivation and then compute the second order interpolating approximations of $\sin(0^{\circ} + \epsilon)$ and $\cos(45^{\circ} + \epsilon)$ for $\epsilon = 10^{-5}$ up to six digits after decimal.	**CO6**	**L4**
12	Give an account of the infinite series representations of several functions proposed by Madhava of Sangamagrama.	**CO6**	**L2**
13	What is the Madhava-Leibnitz series? Discuss the convergence of the series.	**CO6**	**L2**
14	Find the solution of quadratic Diophantine equation by trial and error and using the composition law (Bhavana) : $Nx^2 + C = y^2$ if $N = 2$, $C = 1$	**CO5**	**L2**
15	Find the solution of special equation (Pell's equation) using the composition law and Cakravala algorithm (cyclic method) : $Nx^2 + C = y^2$ if $N = 3$, $C = 1$	**CO5**	**L4**
16	Solve the following equation using Bhavana to transform fractional solutions to integer solutions : $\boldsymbol{92x^2 + 1 = y^2}$	**CO5**	**L4**
17	Solve the following equation using Cakravala rational solutions and scaling method : $\boldsymbol{3x^2 - 800 = y^2}$	**CO5**	**L4**
18	Why is Brahmagupta's formula considered a generalization of Heron's formula?	**CO5**	**L4**
19	Prove Narayana's beautiful area theorem : $\boldsymbol{2AD = d_1 d_2 d_3}$. Why does Narayana's theorem reduce to Brahmagupta's theorem when one diagonal becomes the diameter? Why does Brahmagupta's area formula fail for non-	**CO5**	**L4**

	cyclic quadrilaterals?		
20	Explain the structural difference between Greek and Indian cyclic geometry. Derive Ptolemy’s theorem from Narayana’s theorem.	**CO5**	**L4**
21	Show analytically why a cyclic quadrilateral has only three independent sides and also show analytically how area symmetry appears in cyclic quadrilaterals.	**CO5**	**L4**
22	Explain mathematically how altitude separation measures deviation from Orthogonality and also explain why cyclic quadrilateral theory represents interaction between algebra and geometry.	**CO5**	**L3**
23	Explain the significance of Narayana’s third diagonal. Why was the third diagonal revolutionary mathematically?	**CO5**	**L3**
24	Using Vedic Sutras and meanings solve the followings: a) 'Yaavadunam' : 9692 = ? , 922 = ?, 1042 = ? , 9932 = ? b) 'Ekanyunena Purvena': 9999 × 2378 = ? , 347 × 99999 = ?, 999 × 324567 = ?, 8765 × 9999 = ? c) 'Anurupyena' : 48 × 42 = ? , 46 × 44 = ?, 61 × 43 = ? , 47 × 43 = ? d) 'Dhvajanka' : 38982 ÷ 73 = ? , 7332 ÷ 64 = ?, 601325 ÷ 76 = ?, 37941 ÷ 47 = ? e) 'Paraavartya Yojayet' : 289487 ÷ 13103 = ?, 497342 ÷ 121 = ?, 36520981 ÷ 133 = ?, 136972 ÷ 121 = ? f) 'Urdhva-Tiryagbhyam': 94 × 78 = ?,103 × 105 = ?, 24 × 36 = ?, 252 × 846 = ?	**CO5**	**L3**
	End of the Assignment 3		

Note:

1. Assignment carries a weightage of 10 **marks out of 100**
2. Course outcomes 5 and 6 were covered.

Course Outcomes	CO1	Understand the historical development of mathematics in India.
	CO2	Explain the Indian contribution to number systems and arithmetic.
	CO3	Apply classical Indian methods in algebra and equations.
	CO4	Describe developments in geometry and trigonometry
	CO5	Recognize the invention of calculus in the Kerala school.
	CO6	Analyse the nature of mathematical reasoning and proof in Indian tradition.